

LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 1

Course Code: CSE252

Course Title: INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Course Outcome: CO1

Paper: Sample Subjective Questions

Maximum Marks: 10 x 10 = 100

Unit: 1

Unit Title: Foundations of Artificial Intelligence

Unit Mapped RBT: L2 (Understand)

Total Questions: 10

Suggested Duration: Not timed - practice set

Subjective Questions (10 x 10 = 100 Marks)

Q1. Explain how Artificial Intelligence has evolved from rule-based systems to modern learning systems. (10 Marks)

- (a) Describe what an early rule-based system was and the limitation that held it back.
 - (b) Explain what learning from data supplied that rule-based systems could not, and what it cost in return.
- [L2 - Understand]

Q2. Describe the classification of Artificial Intelligence by capability and by functionality. (10 Marks)

- (a) Name and describe the classes on each of the two axes.
 - (b) Explain which of the classes are represented by systems in use today, and which remain theoretical.
- [L2 - Understand]

Q3. Explain the phases of the Artificial Intelligence development lifecycle. (10 Marks)

- (a) Name the phases in order and state what each one produces for the phase that follows.
 - (b) Explain why the lifecycle returns to an earlier phase rather than ending at deployment.
- [L2 - Understand]

Q4. Describe the properties by which the environment of an intelligent agent is characterised. (10 Marks)

- (a) Explain what observability, determinism, dynamism and the number of agents each describe.
 - (b) Characterise the environment of a warehouse robot on each of the four, giving a reason for each.
- [L2 - Understand]

Q5. Explain what makes an agent rational, and how its performance measure should be written. (10 Marks)

- (a) State what a rational agent is required to do and what its rationality is judged against.
 - (b) Explain, with an example, how a badly written performance measure can be satisfied without the task being done.
- [L2 - Understand]

Q6. Describe the applications of Artificial Intelligence in healthcare and in manufacturing. (10 Marks)

- (a) Describe two applications drawn from each of the two sectors.
 - (b) Explain what the input data looks like in one of them, and one risk the application carries.
- [L2 - Understand]

Q7. Explain the ethical concerns that Artificial Intelligence raises, and the principles of Responsible Artificial Intelligence that address them. (10 Marks)

- (a) Describe three concerns that arise when a system takes decisions about people.
 - (b) Explain which principle addresses each of the three, and one practical step it calls for.
- [L2 - Understand]

Q8. Describe what generative Artificial Intelligence is and explain two of its applications. (10 Marks)

- (a) Describe what distinguishes a generative model from one that classifies or predicts.
 - (b) Explain two applications, and state one risk that follows from the way such a model produces its output.
- [L2 - Understand]

Q9. Explain how bias enters an Artificial Intelligence system and how its effect can be reduced. (10 Marks)

- (a) Describe two routes by which bias reaches a trained model.
- (b) Explain two measures that would reduce the harm, and what each of them leaves unresolved.

[L2 - Understand]

Q10. Distinguish data science from Artificial Intelligence and from machine learning. (10 Marks)

(a) Describe the scope and the purpose of each of the three.

(b) Explain, with one example of each, a task that belongs to data science and one that belongs to Artificial Intelligence.

[L2 - Understand]

----- End of Sample Question Set -----